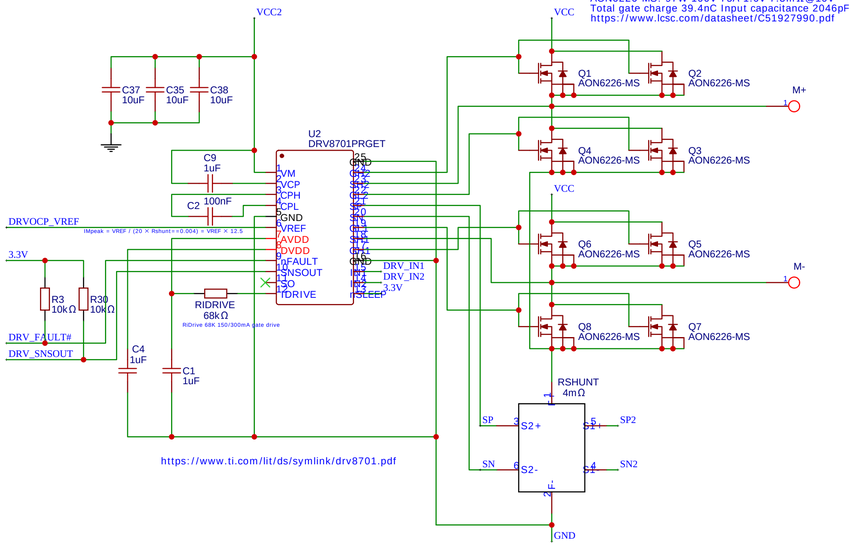
DC Motor Controller with Velocity and PWM Mode

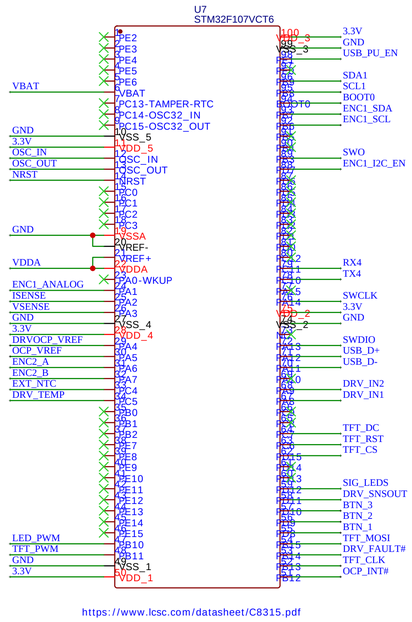
Firmware: https://github.com/sascha432/motor_controller_stm32

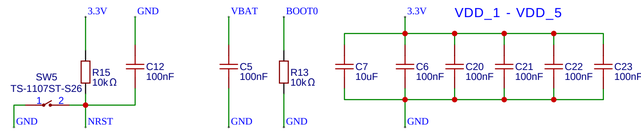
- 8-36V, 20A continuous, 40A peak
- Adjustable cycle by cycle motor current limit (0.5-40A)
- Adjustable max. current with signal LED (0.5-40A)
- 1.14" 135x240 TFT display, rotary encoder and 3 buttons for easy navigation
- Support for magnetic encoder MT6701 in A/B mode up to 1024PPR (55000 RPM)
- Dimmable LED with CC driver 3-32V/350mA/5W
- Current, voltage and temperature monitoring

Motor driver and MOSFETS

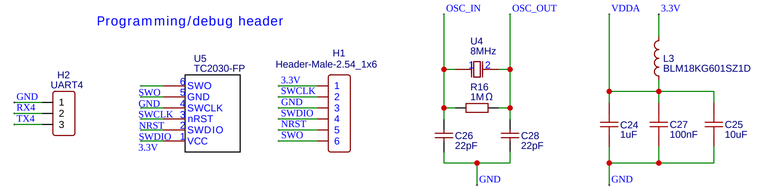


MCU and EEPROM

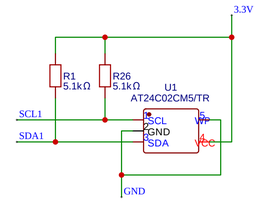




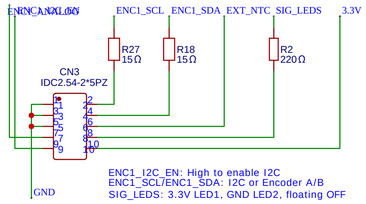
Programming/debug header



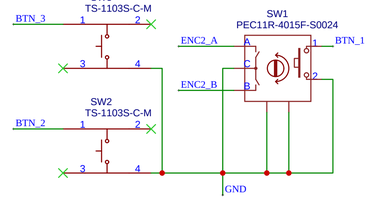
EEPROM and I2C pullups
I2C address 0x50



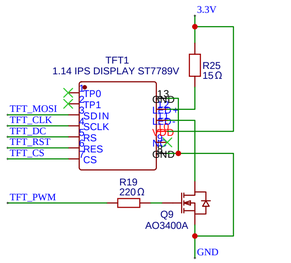
Externals & MT6701 connector
I2C address 0x06



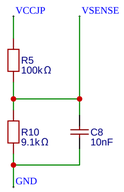
Buttons and rotary encoder



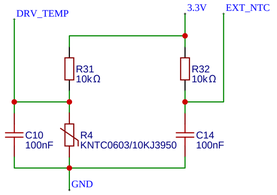
TFT
1.14" 135x240 Full Color IPS Screen 135*240 ST7789 SPI 13 Pins



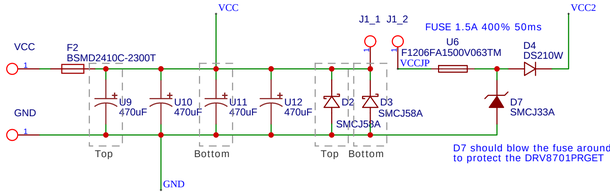
Input voltage detection up to 40V
−3 dB bandwidth ~1.9KHz



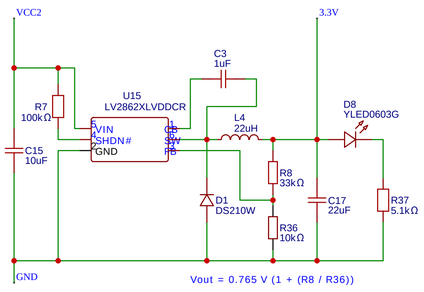
Temperature monitoring



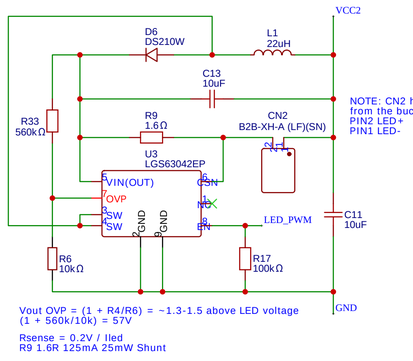
Input bulk capacitance and protection
VCC2 is decoupled from bulk capacitance and fused



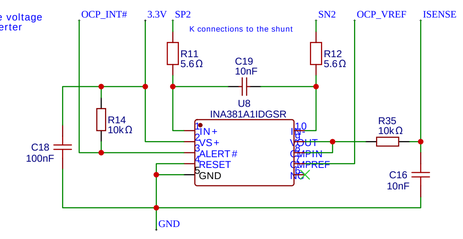
Power Supply 3.3V/0.6A



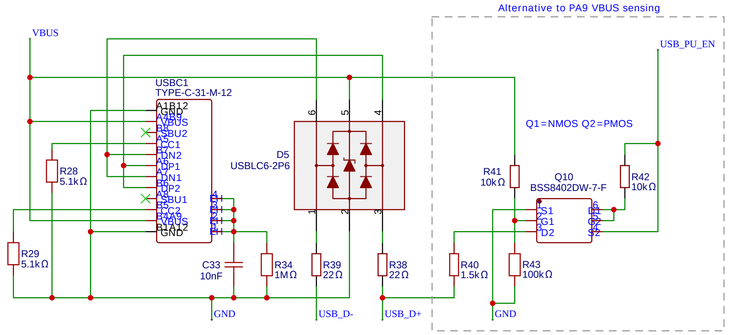
CC LED driver with PWM 2.5-32V/350mA
NOTE: If the LED voltage + VCC exceeds 57V, the OVP is triggered
(OVP - VCC) / 1.3 = max. VLED, (57V - 36V) / 1.3 = ~16V



Input OCP
20x current amplifier with comparator
0.004R shunt = 1A/80mV - 40A/3200mV
Input filter R11, R12 and C19
Output filter R35/C16 for the MCU ADC



USB connector for programming (optional)



Rev1.0 - Initial revision
Rev1.1 - TagConnect footprint with legs, changed C38 to 10uF, power MCU over USB, added header for PC10/11 UART4, added R17 pulldown for LED PWM, fixed K connections to INA381 and a dded RC filter to output, optional power LED D8, pulled DRV8701 nSLEEP to 3.3V, swapped GH1 and GL1, added USB series resistors R38 R39, added USB D+ pullup R40, changed R33 to 560K, changed shunt symbol/footprint
Rev 1.2 - Removed USB LDO (self powered) and added VBUS sensing

TITLE: DC Motor Controller	REV: 1.2
Company: KFCLabs	Sheet: 1/1
Date: 2026-07-28	Drawn By: sascha23095123423

EasyEDA